

Diabetes Prediction - Mini Project 15

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Course: Elements of Applied Artificial Intelligence

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Project Description

This project implements and compares multiple machine learning models to predict diabetes in patients using the **Pima Indians Diabetes dataset**. The models include:

-  **Decision Tree (Default)** - Baseline tree model
-  **Decision Tree (Optimized)** - Tree with GridSearchCV hyperparameter tuning
-  **Logistic Regression** - Linear classification model

The goal is to diagnostically predict whether a patient has diabetes based on diagnostic measurements.

Dataset Information

- **Source:** [Kaggle - Pima Indians Diabetes Database](#) | [UCI Repository](#)
- **Samples:** 768 patients (all females, at least 21 years old, Pima Indian heritage)
- **Features:** 8 diagnostic measurements
- **Target:** Diabetes onset (0 = No, 1 = Yes)

Features:

Feature	Description
Pregnancies	Number of times pregnant
Glucose	Plasma glucose concentration (mg/dL)
BloodPressure	Diastolic blood pressure (mm Hg)
SkinThickness	Triceps skin fold thickness (mm)
Insulin	2-Hour serum insulin (mu U/ml)
BMI	Body mass index (kg/m ²)
DiabetesPedigreeFunction	Diabetes pedigree function (genetic influence)
Age	Age in years
Outcome	1 = Diabetic, 0 = Non-Diabetic

Requirements

```
pip install -r requirements.txt
```

Or install manually:

```
pip install pandas numpy matplotlib seaborn scikit-learn
```

How to Run

Option 1: Run Python Script

```
# Clone the repository
git clone https://github.com/YOUR_USERNAME/diabetes-prediction-ml.git
cd diabetes-prediction-ml

# Run the script
python diabetes_prediction.py
```

Option 2: Jupyter Notebook

You can also run the code in a Jupyter Notebook by copying the contents of `diabetes_prediction.py` into notebook cells.

Results

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
DT (Default)	~0.72	~0.65	~0.58	~0.61	~0.68
DT (Optimized)	~0.75	~0.68	~0.62	~0.65	~0.74
Logistic Regression	~0.77	~0.70	~0.65	~0.67	~0.82

Note: Actual results may vary slightly due to random state and data splits.

Project Structure

```
diabetes-prediction-ml/
├── diabetes.csv           # Dataset (download separately)
├── diabetes_prediction.py # Main Python script
├── requirements.txt       # Python dependencies
├── README.md             # This file
├── 01_distribution.png   # Output: Class distribution chart
├── 02_correlation.png    # Output: Correlation heatmap
├── 03_feature_distributions.png # Output: Feature distributions
├── 04_decision_tree.png  # Output: Decision tree visualization
├── 05_comparison.png     # Output: Model comparison charts
├── 06_confusion_matrices.png # Output: Confusion matrices
└── 07_roc_curves.png     # Output: ROC curves
```

Methodology

1. Data Preprocessing

- Missing Value Handling:** Zero values in biological features (Glucose, BloodPressure, SkinThickness, Insulin, BMI) were replaced with median values since zero is biologically impossible for these measurements.
- Train-Test Split:** 80% training, 20% testing with stratification to maintain class balance.

- **Feature Scaling:** StandardScaler applied for Logistic Regression.

2. Models

- **Decision Tree (Default):** Default scikit-learn parameters (potential overfitting).
- **Decision Tree (Optimized):** GridSearchCV with 5-fold cross-validation optimizing:
 - max_depth: [3, 5, 7, 10, None]
 - min_samples_split: [2, 5, 10, 20]
 - min_samples_leaf: [1, 2, 4, 8]
 - criterion: ['gini', 'entropy']
- **Logistic Regression:** With max_iter=1000 for convergence.

3. Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- AUC-ROC
- Confusion Matrix

🎯 Key Findings

1. **Glucose** is the most important feature for diabetes prediction (highest feature importance).
2. **BMI** and **Age** are also significant predictors.
3. The **Optimized Decision Tree** provides the best balance between interpretability and performance.
4. **Logistic Regression** offers the highest AUC-ROC score, making it suitable for probabilistic medical diagnosis.
5. The **Default Decision Tree** shows signs of overfitting (100% training accuracy vs ~72% test accuracy).

📊 Sample Outputs

Confusion Matrix (Optimized Decision Tree)

		Predicted	
		No Diabetes	Diabetes
Actual	No Diabetes	~85	~15
	Diabetes	~20	~34

Feature Importance Ranking

1. Glucose (~0.30)
2. BMI (~0.18)
3. Age (~0.15)
4. DiabetesPedigreeFunction (~0.12)
5. Pregnancies (~0.10)
6. Insulin (~0.08)
7. BloodPressure (~0.04)

8. SkinThickness (~0.03)

Conclusion

For medical diagnosis applications, the **Optimized Decision Tree** or **Logistic Regression** are recommended due to their:

-  Reliable performance
-  Interpretability
-  Good generalization
-  Probabilistic outputs (Logistic Regression)

The project successfully demonstrates the application of machine learning techniques for diabetes prediction, following the requirements of Mini Project 15.

Contact

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★ **Note:** This project was created as part of the coursework for Elements of Applied Artificial Intelligence at Ferhat Abbas Setif-1 University.